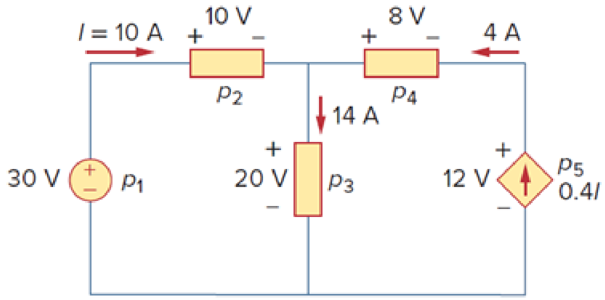
# Chapter 1, Problem 18P

## Problem

Find the power absorbed by each of the elements in Fig. 1.29.

**Figure 1.29:**




## Step-by-step solution

**Step 1** of 4

The following is the given circuit diagram:

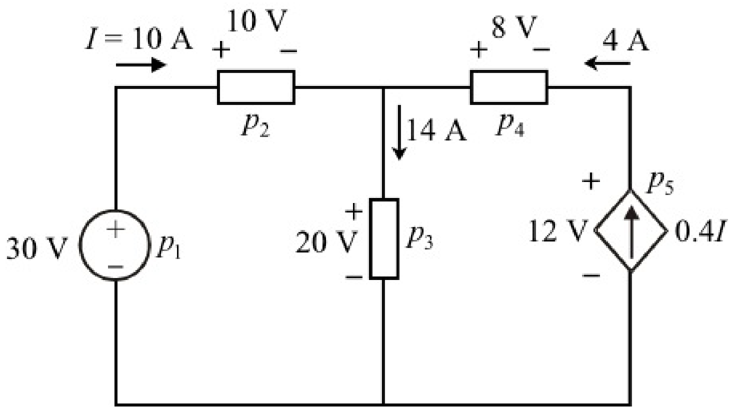



Figure 1

**Step 2** of 4

If the current is entering into positive terminal, the power absorbed is positive $(+\text{ve})$. If the current is entering into negative terminal, the power absorbed is negative $(-\text{ve})$.

The power absorbed by the $30\text{ V}$ source is,

$$\begin{aligned} p_1 &= (-30)(10) \\ &= -300\text{ W} \end{aligned}$$

Therefore, the power absorbed by $30\text{ V}$ source, $p_1$ is $\boxed{-300\text{ W}}$.

The power absorbed by the element 2 is,

$$\begin{aligned} p_2 &= (10)(10) \\ &= 100\text{ W} \end{aligned}$$

Therefore, the power absorbed by element 2, $p_2$ is $\boxed{100\text{ W}}$.

**Step 3** of 4

The power absorbed by the element 3 is,

$$\begin{aligned} p_3 &= (20)(14) \\ &= 280 \text{ W} \end{aligned}$$

Therefore, the power absorbed by element 3, $p_3$ is $\boxed{280 \text{ W}}$.

The power absorbed by the element 4 is,

$$\begin{aligned} p_4 &= (-8)(4) \\ &= -32 \text{ W} \end{aligned}$$

Therefore, the power absorbed by element 4, $p_4$ is $\boxed{-32 \text{ W}}$.

**Step 4** of 4

The power absorbed by the current dependent current source is,

$$\begin{aligned} p_5 &= (-12)(0.4I) \\ &= (-12)(0.4 \times 10) \\ &= -48 \text{ W} \end{aligned}$$

Therefore, the power absorbed by the current dependent current source, $p_5$ is $\boxed{-48 \text{ W}}$.